ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO ROBOTICS

PAPER 1 · PRACTICE D · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Introduction to Robotics paper 1 · practice d.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and wording that shows understanding.

SECTION A

QUESTION 1

1 1.1 C ✓ 1.2 A ✓ 1.3 D ✓ 1.4 B ✓ 1.5 A ✓ 1.6 C ✓ 1.7 B ✓ 1.8 D ✓ 1.9 A ✓ 1.10 B ✓ (10)

[10]

QUESTION 2

2 2.1 True ✓ 2.2 False ✓ 2.3 True ✓ 2.4 False ✓ 2.5 True ✓ (5)

[5]

QUESTION 3

3 3.1 C ✓ 3.2 F ✓ 3.3 B ✓ 3.4 H ✓ 3.5 A ✓ 3.6 E ✓ 3.7 G ✓ 3.8 J ✓ 3.9 I ✓ 3.10 D ✓ (10)

[10]

QUESTION 4

4 4.1 Degrees of Freedom ✓ 4.2 Light-Emitting Diode ✓ 4.3 Internet of Things ✓ 4.4 Universal Serial Bus ✓ 4.5 Artificial Intelligence ✓ (5)

[5]

TOTAL SECTION A: [30]

SECTION B

QUESTION 5

5.1 An autonomous robot senses its surroundings and makes its own decisions without a person controlling it ✓ – e.g. a robot vacuum cleaner, self-driving delivery robot, line follower ✓. A remote-controlled robot is driven by a person through a controller or app ✓ – e.g. a bomb-disposal robot, RC car, drone flown by a pilot ✓. (4)

5.2 A part is designed on a computer (CAD), sliced into layers and printed layer by layer from plastic filament ✓✓. Any TWO advantages ✓✓: parts can be made quickly and cheaply; complex shapes are possible; parts can be changed and reprinted easily; light parts; no special tools needed. (4)

5.3 The trigger sends out a short burst of ultrasonic sound ✓; the sound bounces off the object and returns to the receiver ✓; the sensor measures the time taken ✓; distance = speed of sound × time ÷ 2, because the sound travels there and back ✓. (4)

5.4 Degrees of freedom are the number of independent ways (movements) a robot can move ✓✓; each joint of an arm usually adds one degree of freedom, e.g. a base that rotates, a shoulder, an elbow and a wrist give 4 DOF ✓✓. (4)

5.5 Any FOUR ✓✓✓✓: welding and painting in car factories; mining (inspection, drilling in dangerous areas); agriculture (milking, harvesting, drones spraying crops); medical (surgery robots); warehouses (moving and packing goods); food packaging; bomb disposal / police; cleaning. (4)

[20]

QUESTION 6

6.1 Series: positive of one battery to negative of the next ✓, the voltages add but the capacity stays the same ✓. Parallel: positives joined and negatives joined ✓, the voltage stays the same but the capacity (running time) adds ✓. (4)

6.2 7,4 V ✓; 2 000 mAh ✓ (2)

6.3.1 $P = V \times I = 6 \times 0{,}8 \checkmark = 4{,}8\ W \checkmark$ (2)

6.3.2 $R = V \div I = 6 \div 0{,}8 \checkmark = 7{,}5\ \Omega \checkmark$ (2)

6.4 $d = v \times t \div 2 \checkmark = 343 \times 0{,}0029 \div 2 \checkmark = 0{,}50\ m \checkmark = 49{,}74\ cm \checkmark$ (4)

6.5 Any TWO ✓✓: voltage needed; capacity (running time); maximum current it can supply; mass and size; rechargeable; cost; safety. (2)

[16]

QUESTION 7

7.1 It detects changes in the infrared (heat) radiation given off by a warm body ✓ when the person moves across its field of view ✓. (2)

7.2 The ESP32 has built-in Wi-Fi (and Bluetooth) ✓ to send messages to the phone; it is also faster with more memory for the camera ✓. (2)

7.3 There is too little visible light at night ✓; IR LEDs light up the scene with infrared light that the camera can see but people cannot ✓. (2)

7.4 Any TWO ✓✓: battery capacity large enough for the whole night; a charging / docking station; switching off parts that are not needed (sleep mode); a low-battery warning; efficient motors. (2)

7.5 Any ONE concern with a way to handle it ✓✓, e.g. photos of people taken without consent – put up notices, only store photos when an alarm is triggered, delete them after a set time, keep them secure (POPIA). (2)

[10]

QUESTION 8

8.1 Line 1 includes the Servo library ✓, which has the functions needed to control servo motors ✓. Lines 2 and 3 create two Servo objects ✓, one for each motor, so each can be attached to a pin and given an angle ✓. (4)

8.2 map() changes a number from one range to another ✓ – here the potentiometer reading 0–1023 becomes an angle of 0–180° ✓. 512 × 180 ÷ 1023 ✓ ≈ 90° ✓ (4)

8.3 To stop the arm from hitting its own frame or the table ✓✓ and to protect the servo and the mechanism from moving past their safe limits ✓✓. (4)

8.4 The variable locked changes to the opposite value (true becomes false, false becomes true) ✓✓; the LED shows the new state (on = locked) ✓; while locked, lines 24–27 are skipped, so the arm stays where it is ✓. (4)

8.5 It gives the button time to be released and stops contact bounce ✓✓, so one press is not read many times, which would lock and unlock the arm over and over ✓✓. (4)

8.6 It switches on the internal pull-up resistor ✓, so the input is HIGH when the button is not pressed ✓; the button connects the pin to GND ✓, so pressing it makes the input LOW ✓. (4)

[24]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE